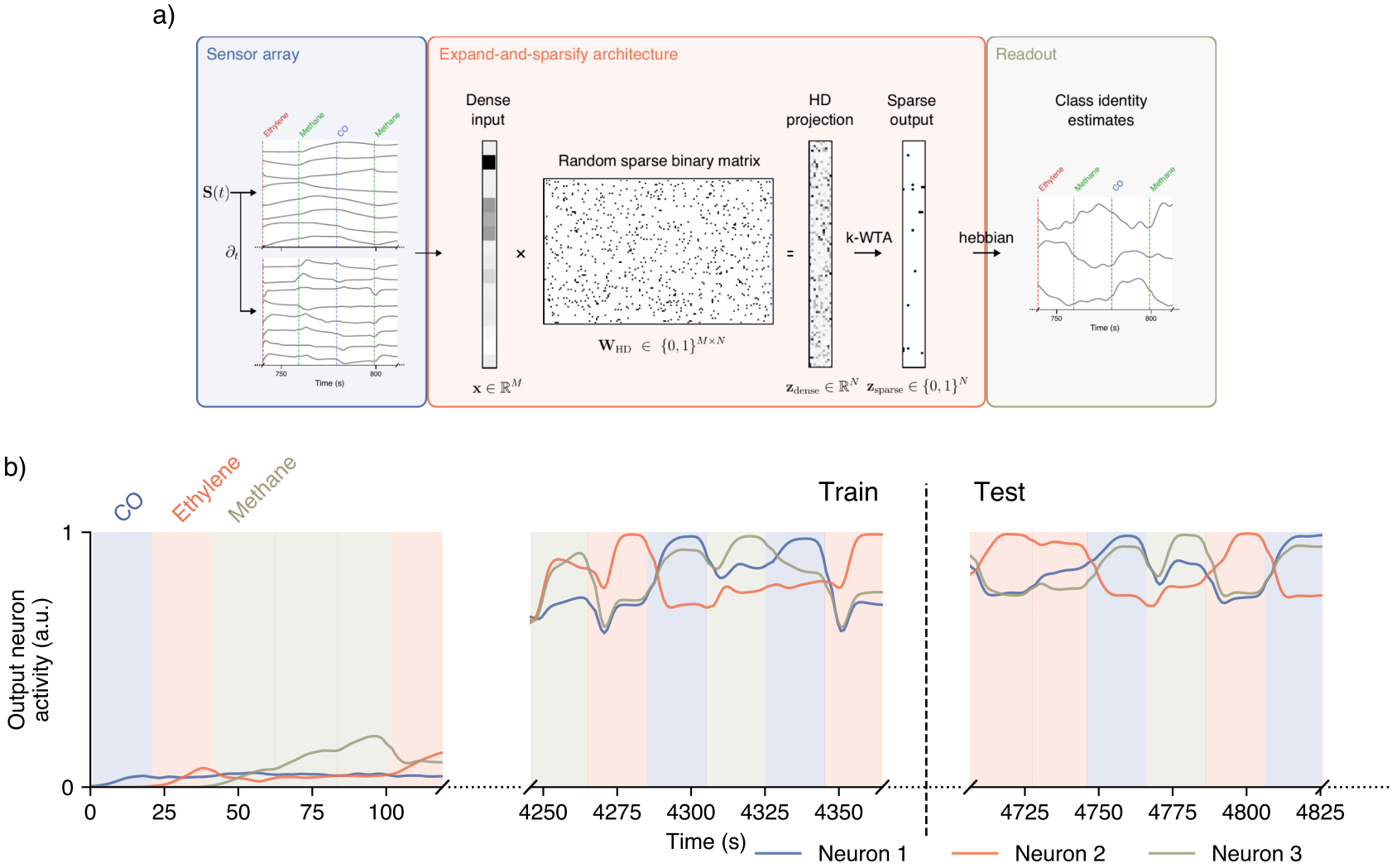
a)
Sensor array
Ethylene
Methane
CO
Methane
S(t)
∂t
750
800
Time (s)
Expand-and-sparsify architecture
Dense input
Random sparse binary matrix
HD projection
Sparse output
k-WTA
x ∈ ℝ^M
W_HD ∈ {0,1}^{M×N}
z_dense ∈ ℝ^N
z_sparse ∈ {0,1}^N
Readout
Class identity estimates
hebbian
b)
CO
Ethylene
Methane
Train
Test
Output neuron activity (a.u.)
1
0
0
25
50
75
100
4250
4275
4300
4325
4350
4725
4750
4775
4800
4825
Time (s)
Neuron 1
Neuron 2
Neuron 3